

Superrigidity

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1 Statement and some Corollaries

Throughout, we assume that $K/\mathbb{k}$ is an extension of fields, with $\mathbb{k}$ a local field of characteristic zero, and K algebraically closed.

Theorem 1.1 (Margulis Superrigidity). *Let $G \leq \mathrm{SL}_n(\mathbb{R})$ be a semisimple algebraic group with $\mathbb{R}\text{-rk}(G) \geq 2$ and no compact factors. Let $\Gamma \leq G$ be an irreducible lattice. Let H be a simple noncompact $\mathbb{R}$ -algebraic group. Suppose $\pi: \Gamma \rightarrow H$ is a homomorphism with $\pi(\Gamma)$ Zariski dense. Then π extends to a rational homomorphism $G \rightarrow H$.*

2 Proof of Margulis Superrigidity for $\mathbb{k} = \mathbb{R}$

Proposition 2.1. *Suppose $P < G$ and $L < H$ are proper algebraic $\mathbb{R}$ -subgroups, and there is a rational Γ -equivariant map $\phi: G/P \rightarrow H/L$ defined over $\mathbb{R}$. Then π extends to a rational homomorphism $G \rightarrow H$.*

Proof. Consider the graph-closure $\mathcal{Z} \subseteq G \times H$ given by

$$\mathcal{Z} = \overline{\{(\gamma, \pi(\gamma)) : \gamma \in \Gamma\}}^{\mathrm{Zar}}.$$

We desire to show that $\mathcal{Z}$ is the graph of a rational homomorphism. To see this, note first that $\Gamma \leq G$ is Zariski dense by Borel density, and $\pi(\Gamma) \leq H$ is dense by assumption. Thus, the projection maps of $\mathcal{Z}$ onto each factor is surjective.

We claim that if $(g, h_1), (g, h_2) \in \mathcal{Z}$, then $h_1 = h_2$. To see this, let $R = \mathrm{Rat}(G/P, H/L)$ be the space of rational maps $G/P \rightarrow H/L$. Then we have an action of $G \times H$ on R given by

$$[(g, h) \cdot \phi](x) = h\phi(g^{-1}x).$$

Suppose $\phi \in R$ is Γ -equivariant. Then

$$[(\gamma, \pi(\gamma)) \cdot \phi](x) = \pi(\gamma)\phi(\gamma^{-1}x) = \pi(\gamma)\pi(\gamma^{-1})\phi(x) = \phi(x),$$

so ϕ is invariant under the action of the graph of π . As ϕ is a rational map, the stabilizer of ϕ is an $\mathbb{R}$ -algebraic subgroup containing the graph of π , thus $\mathcal{Z}$ must stabilize ϕ . In particular, we have

$$h_1\phi(x) = \phi(gx) = h_2\phi(x).$$

Thus, $h_1h_2^{-1}$ leaves $\phi(G/P)$ invariant pointwise, and since it acts algebraically, we have that $h_1h_2^{-1}$ leaves $\overline{\phi(G/P)}^{\mathrm{Zar}}$ invariant pointwise. At the same time, since $\pi(\Gamma)$ leaves $\phi(G/P)$ invariant, it also leaves $\overline{\phi(G/P)}^{\mathrm{Zar}}$ invariant, and since $\pi(\Gamma)$ is Zariski dense in H , we have that H leaves $\overline{\phi(G/P)}^{\mathrm{Zar}}$ invariant, so in fact $\phi(G/P)$ is Zariski dense in H/L , and so $h_1h_2^{-1}$ fixes H/L . But then

$$h_1h_2^{-1} \in \bigcap_{h \in H} hLh^{-1},$$

and since this is a proper normal subgroup of the simple group H , it follows that $h_1 = h_2$.

This implies that the projection map $\mathcal{Z} \rightarrow G$ is in fact a bijective regular map, and so it has a rational inverse $\rho_1: G \rightarrow \mathcal{Z}$. Letting $\rho_2: \mathcal{Z} \rightarrow H$ be the projection map, we see that $\rho_2\rho_1: G \rightarrow H$ is a rational map which extends π . Moreover, it is a homomorphism on a Zariski dense subset, and so it is in fact a homomorphism. $\square$

Now, let $P \leq G$ be a minimal parabolic subgroup, i.e., the intersection in $\mathrm{SL}_n(\mathbb{R})$ of G with the subgroup of upper triangular matrices.

Proposition 2.2. *There exists a proper algebraic $\mathbb{R}$ -subgroup $L < H$ and a measurable Γ -equivariant map $\phi: G/P \rightarrow H/L$.*

Proof. Let $Q < H$ be a proper parabolic subgroup defined over $\mathbb{R}$. Then H/Q is a compact metrizable Γ -space. Consider the product space $G/P \times \mathcal{M}(H/Q)$. Since P is minimal parabolic, P is solvable, hence amenable. As $\Gamma \leq G$ is a closed subgroup, by amenability inheritance there exists a measurable Γ -equivariant section $\phi: G/P \rightarrow \mathcal{M}(H/Q)$.

Since $Q \leq H$ is $\mathbb{R}$ -cocompact, then a result of Zimmer states that H acts smoothly on $\mathcal{M}(H/Q)$, i.e., the quotient $H \backslash \mathcal{M}(H/Q)$ is countably separated. Let $\bar{\phi}: G/P \rightarrow H \backslash \mathcal{M}(H/Q)$ be the induced map. As ϕ is Γ -invariant (on a conull set), we have

$$\bar{\phi}(\gamma x) = \pi(\gamma)\bar{\phi}(x) = \bar{\phi}(x),$$

so $\bar{\phi}$ is constant along Γ -orbits. But the codomain of $\bar{\phi}$ is a countably separated space, and Γ acts on G/P ergodically by the Howe-Moore vanishing theorem. Thus $\bar{\phi}$ is a.e. constant, i.e., ϕ has essential range given by a single H -orbit $H\mu$. But the stabilizer L of μ under the action of H is a proper algebraic $\mathbb{R}$ -subgroup. Again by smoothness, the map $H/L \rightarrow H\mu$ is a homeomorphism, and so we may view ϕ as a Γ -equivariant map $G/P \rightarrow H/L$, as desired. $\square$

Proposition 2.3. *Every measurable Γ -equivariant map $\phi: G/P \rightarrow H/L$ is rational.*

The proof of this proposition is by far the most difficult step in the proof, and proceed via a series of reductions. To start, we will lift ϕ to a right P -invariant map $G \rightarrow H/L$. We also let U denote the intersection of G in $\mathrm{SL}_n(\mathbb{R})$ with the subgroup of unipotent lower triangular matrices.

A rational function $U \rightarrow H/L$ then defines a P -invariant function $U \times P \rightarrow H/L$, which in turn defines a map $G/P \rightarrow H/L$. Thus, if $\phi|_U$ is rational, then ϕ is rational.

We have another black box from Lie theory:

Lemma 2.4. *Suppose G has $\mathbb{R}$ -rank at least 2. There exist nonidentity elements $t_1, \dots, t_n \in A$ and connected algebraic unipotent subgroups U_i so that*

- (i) *for each i , U_i is contained in a semisimple $\mathbb{R}$ -group C_i of elements centralizing t_i ;*
- (ii) *the product map $U_1 \times \dots \times U_n \rightarrow U$ is an $\mathbb{R}$ -isomorphism of varieties;*
- (iii) *for each r , $U_1 \cdots U_r$ is a subgroup of U , and is normal in U_{r+1} .*

In the case of $\mathrm{SL}_3(\mathbb{R})$, we can take for example

$$t_1 = \begin{bmatrix} \lambda & & \\ & \lambda & \\ & & \lambda^{-2} \end{bmatrix}, \quad U_1 = \begin{bmatrix} 1 & & \\ \mathbb{R} & 1 & \\ & & 1 \end{bmatrix}, \quad C_1 = \begin{bmatrix} \mathrm{GL}_2(\mathbb{R}) & \\ & \det^{-1} \end{bmatrix},$$

and similarly for the other elements. Note that this decomposition is not possible for $\mathrm{SL}_2(\mathbb{R})$, and the higher rank condition is essential for the general proof.

Lemma 2.5. *If, for a.e. $g \in G$ and each i , the map $U_i \rightarrow H/L$, $u \mapsto \phi(gu)$ is essentially rational, then $\phi|_U$ is essentially rational.*

Proof. we induct on the following claim: for almost all $g \in G$, the map $u \mapsto \phi(gu)$ is rational on $U_1 \cdots U_r$. When $r = 1$, this is exactly the assumption of the lemma. Suppose then that the statement holds for a.e. $g \in G$ on $U_1 \cdots U_r$. Choose $u \in U_1 \cdots U_r$ and $v \in U_{r+1}$. For each g , consider the function

$$\phi_g(u, v) = \phi(guv).$$

By assumption, for each u , this map is a.e. rational in v , and so by Fubini's theorem, for a.e. g , ϕ_g is rational in v for a.e. u . On the other hand, we have $\phi_g(u, v) = \phi(gv(v^{-1}uv))$, and since v normalizes $U_1 \cdots U_r$, it follows that ϕ_g is also essentially rational in u for a.e. g . But then ϕ_g is essentially rational in both the u and v factor for a.e. g . This in turn implies that ϕ_g is essentially rational on $U_1 \cdots U_{r+1}$, completing the induction. $\square$

Lemma 2.6. *Suppose that for each $1 \leq i \leq n$, there exists a measurable map $h: G \rightarrow H$ (perhaps depending on i) so that $\phi(gc) = h(g)\phi(c)$ for a.e. $c \in C_i$ and $g \in G$, then $\phi|_U$ is rational.*

Proof. Let B denote the subgroup of H fixing C_i pointwise. Note that this subgroup is algebraic, as it is the intersection of stabilizers.

First, we show that $\text{im } h$ normalizes B . Given $b \in B$ and $c, c' \in C_i$, we have

$$[h(c)^{-1}bh(c)]\phi(c') = h(c)^{-1}b\phi(cc') = h(c)^{-1}\phi(cc') = \phi(c'),$$

This shows $\text{im } h$ normalizes B .

Now, consider the induced map $\bar{h}: C_i \rightarrow N_G(B)/B$ given by projection. We claim $\bar{h}$ is a homomorphism. To see this, choose $c, c', c'' \in C_i$. We have

$$h(cc')\phi(c'') = \phi(cc'c'') = h(c)h(c')\phi(c''),$$

and so $h(cc')^{-1}h(c)h(c') \in B$, proving that $\bar{h}$ is a homomorphism.

Finally, recall that C_i is semisimple, so the image of U_i under $\bar{h}$ is unipotent by automatic unipotence, and hence the restriction of $\bar{h}$ to U_i is rational by automatic rationality. Thus, for each $u \in U_i$, we have

$$\phi(gu) = h(g)\phi(u) = h(g)\bar{h}(u)\phi(1).$$

But this map is rational for fixed g , and so $\phi|_{U_i}$ is rational, as desired. $\square$

Proposition 2.7. *Any Γ -equivariant measurable map $\phi: G/P \rightarrow H/L$ is rational.*

Proof. Abusing notation, instead let ϕ denote the P -invariant map $G \rightarrow H/L$. Let $\mathcal{F}(C_i, H/L)$ denote the space of measurable maps $C_i \rightarrow H/L$, modulo a.e. equivalence. Consider the function $\Phi: G \rightarrow \mathcal{F}(C_i, H/L)$ given by

$$\Phi(g)(c) = \phi(gc).$$

Then, for each t_i , as C_i centralizes t_i and ϕ is P -invariant, we have

$$\Phi(gt_i)(c) = \phi(gt_i c) = \phi(gct_i) = \phi(gc) = \Phi(g)(c)$$

for all $c \in C_i$. Let T_i denote the one-parameter subgroup of A containing t_i . As ϕ is Γ -equivariant, the map $\bar{\Phi}: G/T_i \rightarrow H \backslash \mathcal{F}(C_i, H/L)$ is constant along Γ -orbits. But T_i is noncompact, so Γ acts on G/T_i ergodically by Howe-Moore. Moreover, H acts on $\mathcal{F}(C_i, H/L)$ smoothly (since C_i is σ -finite and H/L is an $\mathbb{R}$ -variety on which H acts by regular maps). Thus ergodicity of Γ implies that $\bar{\Phi}$ is essentially constant, i.e., the function

$$\Phi(g)(c) = H\phi(gc)$$

is essentially constant in g . In particular, we must have $\phi(gc) = h(g)\phi(c)$ for some function $h: G \rightarrow H$. This finishes the proof. $\square$

3 Required Results

Theorem 3.1 (Howe-Moore Vanishing Theorem). *Suppose G is a semisimple connected Lie group with finite center and no compact factors. If $H \leq G$ is a closed, noncompact subgroup, and $\Gamma \leq G$ is a lattice, then Γ acts ergodically on G/H .*

Theorem 3.2 (Smoothness of Actions). *Suppose H is $\mathbb{k}$ -group.*

- (i) *Suppose Q is a $\mathbb{k}$ -cocompact algebraic subgroup in H . Then $H_{\mathbb{k}}$ acts smoothly on $\mathcal{M}(H_{\mathbb{k}}/Q_{\mathbb{k}})$.*
- (ii) *Suppose V is a $\mathbb{k}$ -variety, and H acts $\mathbb{k}$ -regularly on V . Then for any σ -finite measure space X , the $H_{\mathbb{k}}$ acts smoothly on $\mathcal{F}(X, V_{\mathbb{k}})$.*

Theorem 3.3 (Borel Density). *Suppose G is a connected semisimple algebraic $\mathbb{R}$ -group with no compact factors. Then any lattice in G is Zariski dense.*

Theorem 3.4 (Amenability Inheritance). *Suppose G is a locally compact group, and $\Gamma, H \leq G$ are closed subgroups. If H is amenable, then the Γ action on G/H is amenable.*

Theorem 3.5 (Automatic Rationality). *Suppose U is an unipotent $\mathbb{R}$ -group and $\pi: U \rightarrow \mathrm{GL}_n(\mathbb{R})$ is a measurable homomorphism with unipotent image. Then π is a regular map.*

Proof. By conjugating (which is a regular map), we may assume U is a subgroup of the group N_m , the unipotent upper triangular matrices in $\mathrm{GL}_m(\mathbb{R})$. If we let $\mathfrak{n}_m$ denote the Lie algebra of N , then the exponential map $\exp: \mathfrak{n}_m \rightarrow N_m$ given by

$$\exp(g) = \sum_{k=0}^{\infty} \frac{g^k}{k!}$$

is in fact a regular map, since $a \in \mathfrak{n}$ is nilpotent (and so the sum is zero after m terms). Moreover, $\exp$ has an inverse $\log: N_m \rightarrow \mathfrak{n}_m$ given by

$$\log(g) = \sum_{k=0}^{\infty} (-1)^k \frac{(g - \mathrm{id})^k}{k!},$$

which is again a regular map. Thus, if $\mathfrak{u}$ is the Lie algebra of U , then $\exp: \mathfrak{u} \rightarrow U$ defines a biregular $\mathbb{R}$ -isomorphism of varieties.

As π is a measurable homomorphism, it is automatically continuous, hence automatically smooth. Let $d\pi: \mathfrak{u} \rightarrow \mathfrak{gl}_n(\mathbb{R})$ denote the derivative of π at the identity; since π has unipotent image, the codomain of $d\pi$ may be taken to be the Lie algebra $\mathfrak{n}_n$. Then we have a commutative diagram

$$\begin{array}{ccc} U & \xrightarrow{\pi} & N_n \\ \log \downarrow & & \uparrow \exp \\ \mathfrak{u} & \xrightarrow{d\pi} & \mathfrak{n}_n \end{array}$$

But $\log$ and $\exp$ are regular and $d\pi$ is linear, hence π is regular. □

Proposition 3.6 (Automatic Unipotence). *Let $G \leq \mathrm{GL}_m(\mathbb{R})$ be a semisimple Lie group and suppose that $\pi: G \rightarrow \mathrm{GL}_n(\mathbb{R})$ is a measurable representaion. If $U \subseteq G$ is a connected unipotent subgroup, then $\pi|_U$ is a unipotent representation, i.e., π has unipotent image.*

Proposition 3.7 (A.e. Rationality). *Suppose $f: \mathbb{R}^m \times \mathbb{R}^n \rightarrow V$ is a measurable function so that, for a.e. $x, y \mapsto f(x, y)$ is essentially rational, and for a.e. $y, x \mapsto f(x, y)$ is essentially rational, then f is essentially rational.*

Proof. It suffices to show the statement for $m = n = 1$, and we may additionally assume that the maps are rational rather than just essentially rational by Fubini's theorem. Suppose we knew that f agreed with a rational function ϕ on a set $A \times \mathbb{R}$, where $A \subseteq \mathbb{R}$ were a set of positive measure. But then

$$0 = \int_{A \times \mathbb{R}} |f - \phi| = \int_{\mathbb{R}} \left[\int_A |f(x, y) - \phi(x, y)| dx \right] dy.$$

But then $f(\cdot, y) = \phi(\cdot, y)$ a.e. on A for a.e. y . As $f(\cdot, y)$ is assumed to be rational, $f(\cdot, y) = \phi(\cdot, y)$ a.e. on $\mathbb{R}$ for a.e. y , which in turn implies $f = \phi$ a.e.

Let $A_{r,s}$ denote the set of points $x_0 \in \mathbb{R}$ so that $f(x_0, \cdot)$ is, in lowest terms, of the form $P(y)/Q(y)$, where P is of degree r and Q is of degree s . Note that one of these sets has positive measure; set $A = A_{r,s}$ with positive measure. Then, for $x \in A$, we have

$$f(x, y) = \frac{y^r + \sum_{i=0}^{r-1} a_i(x) y^i}{\sum_{i=0}^s b_i(x) y^i}.$$

It thus suffices to show that the functions a_i, b_i are essentially rational. However, if we let $\lambda_1, \dots, \lambda_{r+s+1} \in \mathbb{R}$ denote a set such that, for each j , $f(\cdot, \lambda_j)$ is essentially rational; then the a_i, b_i are solutions to the system of equations

$$f_{\lambda_j}(x) = \frac{\lambda_j^r + \sum_{i=0}^{r-1} a_i(x) \lambda_j^i}{\sum_{i=0}^s b_i(x) \lambda_j^i}.$$

This is a system of equations over the field $\mathbb{C}(x)$ with $r + s + 1$ variables and $r + s + 1$ equations, thus it has a unique solution, which is rational in x by Cramer's rule. $\square$

4 Superrigidity over General Fields

Note that the only place in which we used that H was an algebraic $\mathbb{R}$ -group was in the proof of Proposition 2.2., where we used the fact that L was an algebraic $\mathbb{R}$ -group. For general local fields $\mathbb{k}$ this is not true; we instead have the following result:

Proposition 4.1. *Suppose H is a connected $\mathbb{k}$ -group, almost simple over $\mathbb{k}$, and that $Q < H$ is a proper $\mathbb{k}$ -cocompact subgroup. Then for any $\mu \in \mathcal{M}(H_{\mathbb{k}}/Q_{\mathbb{k}})$, either*

- (i) *the stabilizer $(H_{\mathbb{k}})_{\mu}$ of μ in $H_{\mathbb{k}}$ is compact, or;*
- (ii) *there is an algebraic $\mathbb{k}$ -group $L < G$ with $\dim L < \dim G$ and $L_{\mathbb{k}} \supseteq (H_{\mathbb{k}})_{\mu}$.*

Note that condition (ii) is sufficient to conclude the same result as in proposition 2.2, as we then have a continuous Γ -equivariant surjection $H_{\mathbb{k}}\mu \rightarrow H_{\mathbb{k}}/L_{\mathbb{k}}$: by smoothness the natural map $H_{\mathbb{k}}\mu \rightarrow H_{\mathbb{k}}/(H_{\mathbb{k}})_{\mu}$ is a homeomorphism and $L_{\mathbb{k}}$ is a proper closed subgroup containing $(H_{\mathbb{k}})_{\mu}$, and all maps are obviously Γ -equivariant, so we again obtain a map $G/P \rightarrow H_{\mathbb{k}}/L_{\mathbb{k}}$. Thus, either (i) holds or the rest of the proof follows in the same way and we get Margulis superrigidity.

Proposition 4.2. *Suppose Γ is a locally compact group, (X, μ) is a quasi-invariant Γ -space with quasi-invariant measure, and the action of Γ on X is weak mixing. Let $\pi: \Gamma \rightarrow H$ be a homomorphism with H also locally compact, and suppose there exists a measurable Γ -equivariant map $\phi: X \rightarrow H/K$, where $K \leq H$ is a compact subgroup. Then $\pi(\overline{\Gamma})$ is compact.*

Proof. Without loss of generality, we may choose μ to be a probability measure on X , so that $\nu = \phi_*\mu$ is a probability measure on H/K . The map $\phi \times \phi: X \times X \rightarrow H/K \times H/K$ is still Γ -equivariant; moreover, since the action of Γ is weak mixing, $\mu \times \mu$ is ergodic, and so $\nu \times \nu$ is ergodic. Now, the action of K on H is smooth by compactness of K , and hence the action of H on $H/K \times H/K$ is smooth. Therefore, the induced map $\bar{\phi}: X \times X \rightarrow H \backslash (H/K \times H/K)$ is Γ -invariant, and since the quotient space is countably separated, it follows

that the map is essentially constant, and so $\nu \times \nu$ is supported on an H -orbit $\mathcal{O} = H(u, v)$. By Fubini's theorem, there must exist $x \in H$ so that ν is supported on the slice $E_x = \{yK \in H/K : (xK, yK) \in \mathcal{O}\}$.

Suppose $yK, y'K \in E_x$. Then there exist $h, h' \in H$ so that

$$(xK, yK) = (huK, hvK) \quad \text{and} \quad (xK, y'K) = (h'uK, h'vK).$$

Thus $h^{-1}h' \in uKu^{-1}$, and so there exists $k \in K$ so that $yK = hvK = uku^{-1}h'vK = uku^{-1}y'K$. In particular, the set E_x is a uKu^{-1} -orbit in H/K ; since K is compact, so is uKu^{-1} , and orbits of compact groups are compact, so $E_x \subseteq H/K$ is compact. In other words, $\text{supp } \nu$ is compact. But ν is quasi-invariant with respect to $\pi(\Gamma)$, and so $\pi(\Gamma)$ leaves $\text{supp } \nu$ invariant. Lifting to $\overline{H}$, we see that $\pi(\Gamma)$ leaves a compact set $B \subseteq H$ invariant under translation. Thus $\pi(\Gamma) \subseteq BB^{-1}$, and so $\pi(\Gamma)$ is compact. $\square$

Theorem 4.3 (The Case $\mathbb{k} = \mathbb{C}$). *Let $G \leq \text{SL}_n(\mathbb{R})$ be a semisimple algebraic group with $\mathbb{R} - \text{rk}(G) \geq 2$ and no compact factors. Let $\Gamma \leq G$ be an irreducible lattice. Let H be a simple noncompact $\mathbb{C}$ -algebraic group. Suppose $\pi: \Gamma \rightarrow H_{\mathbb{C}}$ is a homomorphism with $\pi(\Gamma)$ Zariski dense. Then either $\overline{\pi(\Gamma)} \leq H_{\mathbb{C}}$ is compact in the Hausdorff topology or π extends to a rational homomorphism $G \rightarrow H_{\mathbb{C}}$.*

Proof. By Howe-Moore, the action of Γ on G/P is weak mixing. Suppose condition (i) holds in Proposition 4.1. As in the proof of Proposition 2.2., we obtain a measurable Γ -equivariant map $\phi: G/P \rightarrow H_{\mathbb{C}}/(H_{\mathbb{C}})_{\mu}$. By the assumption that the stabilizer of μ is compact, Proposition 4.2 states that $\pi(\Gamma)$ is compact. $\square$

Theorem 4.4 (The Case of $\mathbb{k}$ Nonarchimedean). *Let $G \leq \text{SL}_n(\mathbb{R})$ be a semisimple algebraic group with $\mathbb{R} - \text{rk}(G) \geq 2$ and no compact factors. Let $\Gamma \leq G$ be an irreducible lattice. Let H be a simple noncompact $\mathbb{k}$ -algebraic group, for $\mathbb{k}$ a non-Archimedean local field. Suppose $\pi: \Gamma \rightarrow H_{\mathbb{k}}$ is a homomorphism with $\pi(\Gamma)$ Zariski dense. Then $\overline{\pi(\Gamma)} \leq H_{\mathbb{k}}$ is compact.*

Proof. By Propositions 4.1 and 4.2, it suffices to show that (ii) cannot occur. Suppose for sake of contradiction that, as in Proposition 4.1, there were to exist an algebraic $\mathbb{k}$ -group $L < G$ with $\dim L < \dim G$ such that $L_{\mathbb{k}} \subseteq (H_{\mathbb{k}})_{\mu}$. Then as in Proposition 2.2, there exists a Γ -invariant measurable map $\phi: G/P \rightarrow H/L$. The rest of the proof of superrigidity does not depend on the local field $\mathbb{k}$ that is chosen; at the same time, the maps $\phi|_{U_i}$ as in the proof are then continuous maps from a connected set into a totally disconnected set, thus they must be constant. It then follows that $\phi|_U$ is constant, hence after identifying U with G/P , we see that the rational map ϕ we have constructed is in fact constant. However, Γ -equivariance implies that $\pi(\Gamma)$ is contained in some conjugate of $L_{\mathbb{k}}$, which contradicts Zariski density. $\square$